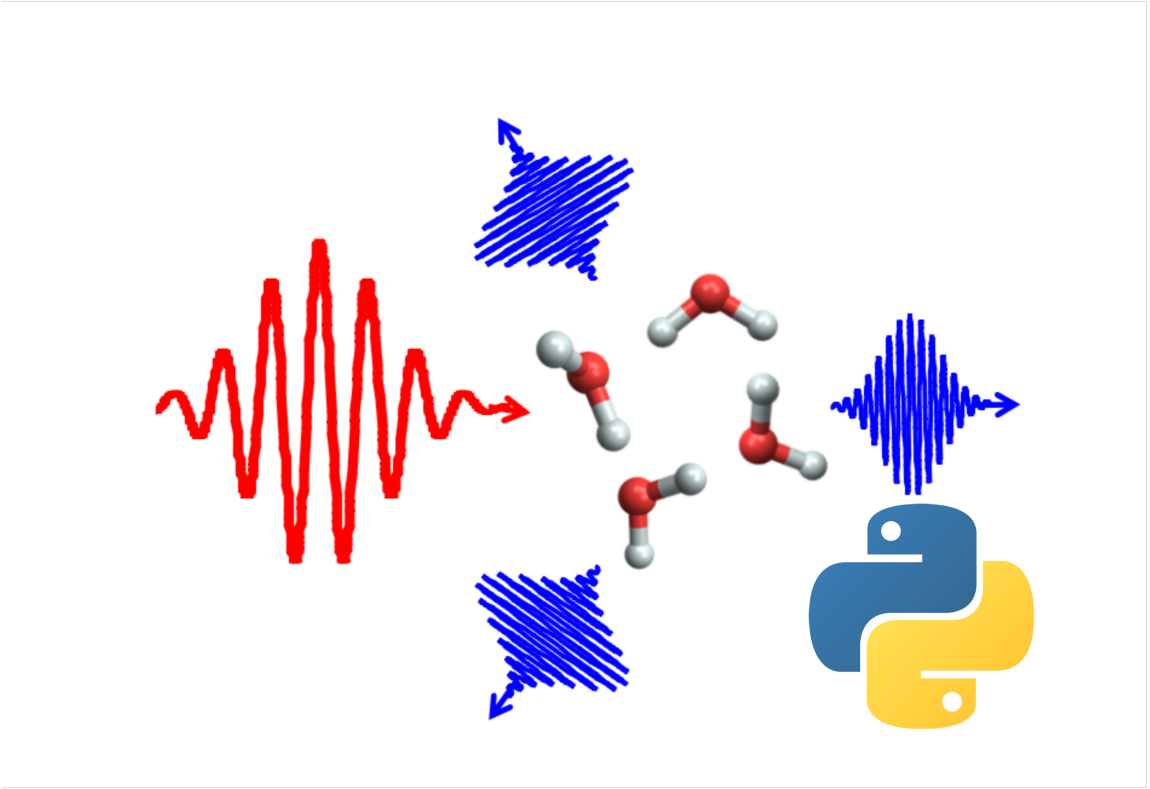
PySHS

A Computational Second Harmonic Scattering Code

User Manual — Version 3.2.3

Scientific documentation & Graphical User Interface



Pierre-Marie Gassin
Institut Charles Gerhardt Montpellier – ENSCM
pierre-marie.gassin@enscm.fr

Contents

1 Introduction **3**
- 1.1 What is PySHS? . . . 3
- 1.2 Distribution and package layout . . . 3
- 1.3 Compilation and installation . . . 4
  - 1.3.1 Requirements (all platforms) . . . 4
  - 1.3.2 Linux / macOS . . . 4
  - 1.3.3 Windows . . . 4
- 1.4 Command-line usage (without GUI) . . . 5

2 General equations implemented in PySHS **6**
- 2.1 Notation . . . 6
- 2.2 Incoherent Second Harmonic Scattering (HRS) . . . 7
- 2.3 Coherent SHS of N fully correlated molecules . . . 9
- 2.4 Coherent SHS by a sphere . . . 11
- 2.5 Coherent SHS by a cylinder . . . 12
  - 2.5.1 Physical model . . . 12
  - 2.5.2 Geometry and local surface frame . . . 13
  - 2.5.3 Effective nonlinear susceptibility . . . 13
  - 2.5.4 Numerical quadrature and orientational average . . . 14
  - 2.5.5 Polarization-resolved coefficients . . . 14

3 HRS program **15**
- 3.1 Inputs and outputs . . . 15
- 3.2 Command line . . . 15

4 SHS program **16**
- 4.1 Inputs . . . 16
- 4.2 Command line . . . 16

5 Sphere Object (`sphere_SHS`) **17**
- 5.1 Inputs . . . 17
- 5.2 Command line . . . 17

6 Cylinder Object (`cylinder_SHS`) **18**
- 6.1 Inputs . . . 18
- 6.2 Command line . . . 18

7 Graphical User Interface **19**
- 7.1 Launch . . . 19
- 7.2 Common layout . . . 19
- 7.3 HRS tab . . . 19
- 7.4 SHS tab . . . 19
- 7.5 Sphere Object and Cylinder Object tabs . . . 20

7.6 Results and export 20

8 Additional scripts **21**
8.1 Orientation / position generators (SHS) 21
8.2 Gaussian → PySHS tensor 21
8.3 Plotting 21

Chapter 1

Introduction

1.1 What is PySHS?

PySHS evaluates the second-harmonic scattered intensity for four classes of systems:

Module	Binary	Physical situation
HRS	`HRS`	Incoherent SHS of uncorrelated molecules (Hyper-Rayleigh Scattering)
SHS	`SHS`	Coherent SHS of N fully correlated molecules (aggregate)
Sphere Object	`sphere_SHS`	Continuum surface response of a colloidal sphere (under the NLWKB theory)
Cylinder Object	`cylinder_SHS`	Continuum surface response of a colloidal cylinder (under the NLWKB theory)

For each module three calculation modes (keywords) are available:

`polarplot_single`
: Polarization-resolved intensity $I_V(\gamma)$, $I_H(\gamma)$ at one fixed scattering angle Θ.

`polarplot_integrate`
: Same polarization plot, coefficients averaged between two scattering angles.

`angle_scattering`
: Intensity versus scattering angle for fixed polarization combinations IV(0°), IV(90°), IH(0°), IH(90°).

1.2 Distribution and package layout

The C cores require compilation. The GUI and helper scripts are written in Python. After unpacking, the parent directory contains:

```
PySHS_GUI/
 run_gui.py              # Launch the GUI (recommended)
 README.md
 Src/                    # All source code
    pyshs_gui.py
    HRS.c, SHS.c, sphere_SHS.c, cylinder_SHS.c
    graph_*.py, script_*.py
```

```
   script_gauss2PySHS.py
bin/                    # Compiled C executables (created automatically)
   HRS, SHS, sphere_SHS, cylinder_SHS   # Linux / macOS
   *.exe                                # Windows
Work/                   # Input / output of calculations only
Documentation/          # User manuals (pdf, tex, docx)
```

Src/ Source code (C modules, GUI, plotting and generation scripts).

bin/ Compiled executables. Created/updated automatically when the GUI starts.

Work/ Working directory for input tensors, orientation files and output results.

Documentation/ This manual and related documents.

1.3 Compilation and installation

1.3.1 Requirements (all platforms)

- Python 3.8+ with `tkinter`, `matplotlib`, `numpy`
- A C compiler: `gcc` or `clang` (on Windows: MSYS2/MinGW-w64 or WinLibs)

1.3.2 Linux / macOS

```
# Optional system packages (Debian/Ubuntu example)
sudo apt install python3-tk python3-matplotlib python3-numpy gcc

# Launch (auto-compiles C sources into bin/)
python3 run_gui.py
```

Manual compilation (optional):

```
gcc -o bin/HRS         Src/HRS.c         -lm
gcc -o bin/SHS         Src/SHS.c         -lm
gcc -o bin/sphere_SHS  Src/sphere_SHS.c  -lm
gcc -o bin/cylinder_SHS Src/cylinder_SHS.c -lm
```

1.3.3 Windows

1. Install Python 3 from https://www.python.org (tick “Add Python to PATH”).
2. Install a MinGW-based toolchain, e.g. MSYS2 UCRT64 (https://www.msys2.org) or WinLibs.
3. From a terminal where `gcc` and `python` are on `PATH`:

   ```
   python run_gui.py
   ```

   or double-click `compile_and_run.bat`.

The GUI compiles the four C modules into `bin/` at startup when a compiler is available.

1.4 Command-line usage (without GUI)

```
# HRS
./bin/HRS <keyword> <input_beta> <output_file>

# SHS (two input files)
./bin/SHS <keyword> <input_beta> <input_orientation> <output_file>

# Sphere / Cylinder (surface chi(2) tensor)
./bin/sphere_SHS  <keyword> <input_beta> <output_file>
./bin/cylinder_SHS <keyword> <input_beta> <output_file>
```

where `<keyword>` is one of `polarplot_single`, `polarplot_integrate`, `angle_scattering`. Additional numerical parameters (wavelength, refractive indices, angles,...) are requested interactively on standard input.

Plotting helpers:

```
python3 Src/graph_polar.py   Work/out_plot
python3 Src/graph_angle.py   Work/out_plot
python3 Src/graph_assembly.py Work/orientation_file
```

Chapter 2

General equations implemented in PySHS

2.1 Notation

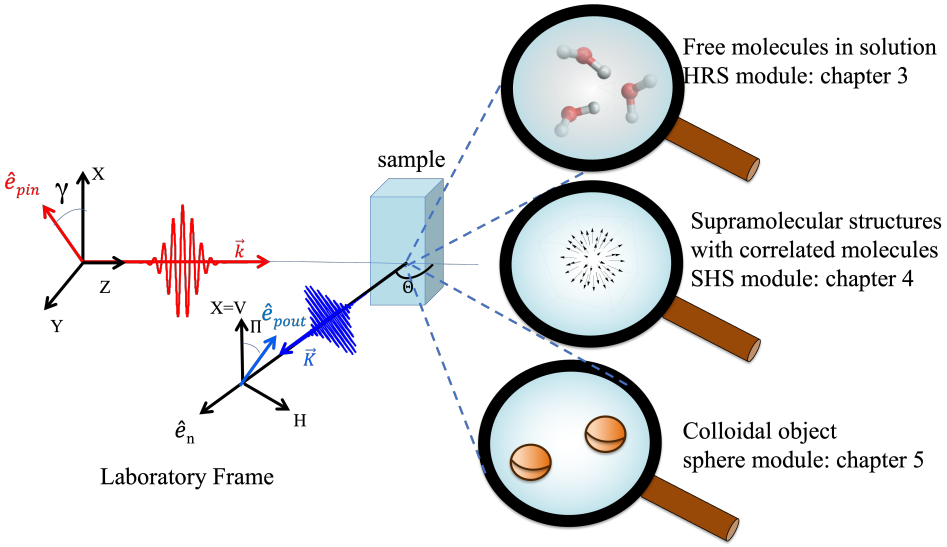


Figure 2.1: Laboratory frame: fundamental wave vector $\mathbf{k}^{\omega}$ and scattered wave vector $\mathbf{K}^{2\omega}$ at angle Θ .

The fundamental field at frequency ω is characterized by its wave vector $k^{\omega}.\mathbf{e}_Z$ and polarization direction $\mathbf{e}^{\omega}_{p-in}$. The input polarization angle γ is the angle between $\mathbf{e}^{\omega}_{p-in}$ and the laboratory X axis. The second-harmonic field at 2ω is characterized by $K^{2\omega}.\mathbf{e}_n$ and the scattering angle Θ. The output polarization angle Π defines the detected polarization $\mathbf{e}^{2\omega}_{p-out}$. In the laboratory frame, the following unit vectors are defined by:

$$\mathbf{e}^{2\omega}_{n} = (0, \sin\Theta,\ \cos\Theta) \tag{2.1}$$

$$\mathbf{e}^{\omega}_{p-in} = (\cos\gamma,\ \sin\gamma,\ 0) \tag{2.2}$$

$$\mathbf{e}^{2\omega}_{p-out} = (\cos\Pi,\ \sin\Pi\cos\Theta,\ \sin\Pi\sin\Theta) \tag{2.3}$$

The scattered wave vector is defined by:

$$\Delta\mathbf{k} = 2\mathbf{k}^{\omega} - \mathbf{K}^{2\omega} = (0, -K\sin\Theta, 2k - K\cos\Theta). \tag{2.4}$$

2.2 Incoherent Second Harmonic Scattering (HRS)

The medium is treated as homogeneous with a single (possibly complex) refractive index n. Wave-vector magnitudes are

$$k^{\omega} = \frac{2\pi n}{\lambda}, \qquad k^{2\omega} = \frac{4\pi n}{\lambda}. \tag{2.5}$$

The Second Harmonic Scattering of uncorrelated molecule in solution is usually referred to Hyper Rayleigh Scattering (HRS) and can be expressed as:

$$I_{HRS}(\mathbf{e}^{\omega}_{p-in}, \mathbf{e}^{2\omega}_{p-out}, \mathbf{e}^{2\omega}_{n}) = I_{HRS}(\gamma, \Pi, \Theta) =< \boldsymbol{\beta}_{eff}(\gamma, \Pi, \Theta).\boldsymbol{\beta}^{*}_{eff}(\gamma, \Pi, \Theta) > \tag{2.6}$$

Here

$$\boldsymbol{\beta_{eff}} = \mathbf{e}^{2\omega}_{n} \times (\overset{\leftrightarrow}{\beta} : \mathbf{e}^{\omega}_{p-in}\mathbf{e}^{\omega}_{p-in}) \times \mathbf{e}^{2\omega}_{n} \tag{2.7}$$

where $\overset{\leftrightarrow}{\beta}$ is the second order hyperpolarisability of the molecule expressed in the laboratory frame. The link between the expression of $\overset{\leftrightarrow}{\beta}$ in the laboratory and microscopic frames is:

$$\beta^{\text{lab}}_{IJK} = \sum_{i=1}^{3}\sum_{j=1}^{3}\sum_{k=1}^{3} T_{Ii}T_{Jj}T_{Kk}\,\beta^{\text{micro}}_{ijk}, \tag{2.8}$$

where β^{lab}_{IJK} is the molecular hyperpolarizability expressed in the laboratory frame and $\beta^{\text{micro}}_{ijk}$ is the molecular hyperpolarizability expressed in the molecular frame. Transformation of the molecular frame uses the elements of the rotation matrix $T_{ab}(\varphi, \theta, \psi)$ using the ZYZ convention:

$$T(\varphi,\theta,\psi) = \begin{pmatrix} \cos\theta\cos\varphi\cos\psi - \sin\varphi\sin\psi & -\cos\psi\sin\varphi - \cos\theta\cos\varphi\sin\psi & \cos\varphi\sin\theta \\ \cos\theta\cos\psi\sin\varphi + \cos\varphi\sin\psi & \cos\varphi\cos\psi - \cos\theta\sin\varphi\sin\psi & \sin\theta\sin\varphi \\ -\cos\psi\sin\theta & \sin\theta\sin\psi & \cos\theta \end{pmatrix} \tag{2.9}$$

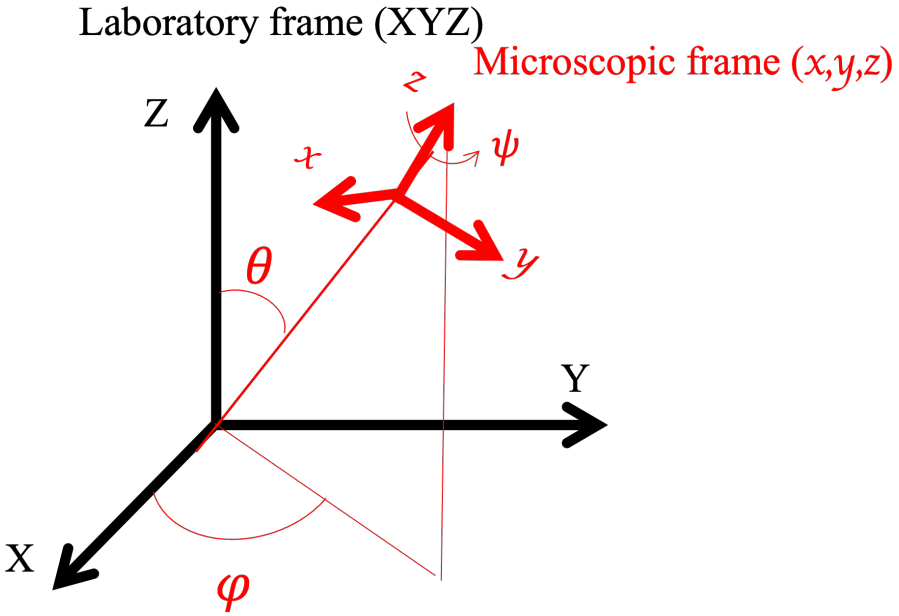


Figure 2.2: Euler angle (φ, θ, ψ) relating the molecular frame to the laboratory frame (ZYZ convention).

Finally, the bracket $\langle\cdot\rangle$ denote the isotropic orientational average.

$$\langle\cdot\rangle = \frac{1}{8\pi^2}\int_{\psi=0}^{2\pi}\int_{\theta=0}^{\pi}\int_{\varphi=0}^{2\pi}\cdot\sin\theta\, d\varphi d\theta\, d\psi \tag{2.10}$$

Expanding (2.6) yields the following expression of the effective hyperpolarizability expressed in term of the tensor components expressed in the laboratory frame:

$$\boldsymbol{\beta_{eff}}(\gamma,\Theta) = \begin{bmatrix} \beta_{XXX}\cos^2\gamma + 2\beta_{XXY}\cos\gamma\sin\gamma + \beta_{XYY}\sin^2\gamma \\ \left(\beta_{YYY}\cos^2\gamma + 2\beta_{YXY}\cos\gamma\sin\gamma + \beta_{YYY}\sin^2\gamma\right)\cos^2\Theta \\ -\left(\beta_{ZYY}\cos^2\gamma + 2\beta_{ZXY}\cos\gamma\sin\gamma + \beta_{ZYY}\sin^2\gamma\right)\cos\Theta\sin\Theta \\ -\left(\beta_{YYY}\cos^2\gamma + 2\beta_{YXY}\cos\gamma\sin\gamma + \beta_{YYY}\sin^2\gamma\right)\cos\Theta\sin\Theta \\ +\left(\beta_{ZYY}\cos^2\gamma + 2\beta_{ZXY}\cos\gamma\sin\gamma + \beta_{ZYY}\sin^2\gamma\right)\sin^2\Theta \end{bmatrix} \tag{2.11}$$

The equations (2.6) and (2.11) can be expressed as polarization resolved plots:

$$I_V(\gamma,\Theta) = a_V^\Theta\cos^4\gamma + b_V^\Theta\cos^2\gamma\sin^2\gamma + c_V^\Theta\sin^4\gamma, \tag{2.12}$$

$$I_H(\gamma,,\Theta) = a_H^\Theta\cos^4\gamma + b_H^\Theta\cos^2\gamma\sin^2\gamma + c_H^\Theta\sin^4\gamma. \tag{2.13}$$

The six coefficients $a_V^\Theta, b_V^\Theta, c_V^\Theta, a_H^\Theta, b_H^\Theta, c_H^\Theta$ are combinations of orientationally averaged products of laboratory-frame β components according to:

$$a_V^\Theta = \langle\beta_{XXX}\beta^*_{XXX}\rangle \tag{2.14}$$

$$c_V^\Theta = \langle\beta_{XYY}\beta^*_{XYY}\rangle \tag{2.15}$$

$$b_V^\Theta = \langle 4\beta_{XXY}\beta^*_{XXY} + \beta_{XXX}\beta^*_{XYY} + \beta^*_{XXX}\beta_{XYY}\rangle \tag{2.16}$$

$$a_H^\Theta = \langle\beta_{YXX}\beta^*_{YXX}\cos^2\rangle\Theta + \langle\beta_{ZXX}\beta^*_{ZXX}\rangle\sin^2\Theta - \langle\beta_{YXX}\beta^*_{ZXX} + \beta_{ZXX}\beta^*_{YXX}\rangle\cos\Theta\sin\Theta \tag{2.17}$$

$$c_H^\Theta = \langle\beta_{YYY}\beta^*_{YYY}\cos^2\rangle\Theta + \langle\beta_{ZYY}\beta^*_{ZYY}\rangle\sin^2\Theta - \langle\beta_{YYY}\beta^*_{ZYY} + \beta_{ZYY}\beta^*_{YYY}\rangle\cos\Theta\sin\Theta \tag{2.18}$$

$$\begin{aligned} b_H^\Theta = {} & \left\langle 4\beta_{ZXY}\beta^*_{ZXY} + \beta_{ZXX}\beta^*_{ZYY} + \beta^*_{ZXX}\beta_{ZYY}\right\rangle\sin^2\Theta \\ & + \left\langle 4\beta_{YYX}\beta^*_{YYX} + \beta_{YXX}\beta^*_{YYY} + \beta^*_{YXX}\beta_{YYY}\right\rangle\cos^2\Theta \\ & - \cos\Theta\sin\Theta\big\langle 4\beta_{YYX}\beta^*_{ZXY} + 4\beta_{ZXY}\beta^*_{YYX} + \beta_{ZXX}\beta^*_{YYY} + \beta^*_{ZXX}\beta_{YYY} \\ & \quad + \beta_{YXX}\beta^*_{ZYY} + \beta_{ZYY}\beta^*_{YXX}\big\rangle \end{aligned} \tag{2.19}$$

Additional dimensionless ratios $I_2^{\Theta,\Pi}$ and $I_4^{\Theta,\Pi}$ can also be computed:

$$I_2^{\Theta,\Pi} = \frac{4(a_\Pi^\Theta - c_\Pi^\Theta)}{3a_\Pi^\Theta + b_\Pi^\Theta + 3c_\Pi^\Theta}, \qquad I_4^{\Theta,\Pi} = \frac{a_\Pi^\Theta - b_\Pi^\Theta + c_\Pi^\Theta}{3a_\Pi^\Theta + b_\Pi^\Theta + 3c_\Pi^\Theta}, \tag{2.20}$$

$$\tag{2.21}$$

In `angle_scattering` mode the program evaluates the following quantities as functions of the scattering angle Θ.

$$I_V(0^\circ) = I_{HRS}(\gamma=0,\Pi=0,\Theta) = a_V^\Theta; \tag{2.22}$$

$$I_V(90^\circ) = I_{HRS}(\gamma=90,\Pi=0,\Theta) = c_V^\Theta, \tag{2.23}$$

$$I_H(0^\circ) = I_{HRS}(\gamma=0,\Pi=90,\Theta) = a_H^\Theta, \tag{2.24}$$

$$I_H(90^\circ) = I_{HRS}(\gamma=90,\Pi=90,\Theta) = c_H^\Theta \tag{2.25}$$

The program can also report the isotropic HRS hyperpolarizability defined for $\Theta = 90$:

$$\beta_{\text{HRS}} = \sqrt{\langle\beta^2_{XXX}\rangle + \langle\beta^2_{ZXX}\rangle} \tag{2.26}$$

and the depolarization ratio

$$D = \frac{c_V^\Theta}{a_V^\Theta} = \frac{\langle\beta^2_{XYY}\rangle}{\langle\beta^2_{XXX}\rangle}. \tag{2.27}$$

2.3 Coherent SHS of N fully correlated molecules

The medium is again considered homogeneous with refractive index n and wave vectors (2.5). The coherent intensity of N fully correlated molecules is given by:

$$I_{SHS}(\mathbf{e}^{\omega}_{p-in}, \mathbf{e}^{2\omega}_{p-out}, \mathbf{e}^{2\omega}_{n}) = I_{SHS}(\gamma, \Pi, \Theta) =< \boldsymbol{\beta}_{eff,t}(\gamma, \Pi, \Theta).\boldsymbol{\beta}^{*}_{eff,t}(\gamma, \Pi, \Theta) > \tag{2.28}$$

where $\boldsymbol{\beta}_{eff,t}$ is defined as:

$$\boldsymbol{\beta}_{eff,t} = \mathbf{e}^{2\omega}_{n} \times (\overleftrightarrow{\beta}_{\mathbf{t}} : \mathbf{e}^{\omega}_{p-in}\mathbf{e}^{\omega}_{p-in}) \times \mathbf{e}^{2\omega}_{n} \tag{2.29}$$

here

$$\overleftrightarrow{\beta}_{\mathbf{t}} = \sum_{l=0}^{N} \overleftrightarrow{\beta}_{l} e^{i\Delta\mathbf{k}.\mathbf{r}_l} \tag{2.30}$$

and $\overleftrightarrow{\beta}_{j}$ is the second order hyperpolarizability tensor expressed in the laboratory frame of the molecule l located at the position $\boldsymbol{r}_l$. From the position $\boldsymbol{r}_l = (x_l, y_l, z_l)$ and the orientation $\Omega_l = (\varphi_l, \theta_l, \psi_l)$ of the molecule l expressed in the mesoscopic (or particule) frame, the quantities $\mathbf{r}^{Labo}_{l}$ and β^{Labo}_{l} are expressed according:

$$\mathbf{r}^{Lab}_{l,I} = \sum_{i'=1}^{3} T_{Ii'}(\varphi, \theta, \psi)\mathbf{r}^{meso}_{l,i'} \tag{2.31}$$

here $\mathbf{r}^{meso}_{l,i'}$ represent the i'coordinates in the mesoscopic frame of the molecule l.

$$\beta^{\text{lab}}_{l,IJK} = \sum_{i'=1}^{3}\sum_{j'=1}^{3}\sum_{k'=1}^{3} T_{Ii}(\varphi_l, \theta_l, \psi_l)T_{Jj}(\varphi_l, \theta_l, \psi_l)T_{Kk}(\varphi_l, \theta_l, \psi_l)\, \beta^{\text{meso}}_{l,i'j'k'} \tag{2.32}$$

$$\beta^{\text{meso}}_{l,i'j'k'} = \sum_{i=1}^{3}\sum_{j=1}^{3}\sum_{k=1}^{3} T_{i'i}(\varphi'_l, \theta'_l, \psi'_l)T_{j'j}(\varphi'_l, \theta'_l, \psi'_l)T_{k'k}(\varphi'_l, \theta'_l, \psi'_l)\, \beta^{\text{molecular}}_{l,ijk}, \tag{2.33}$$

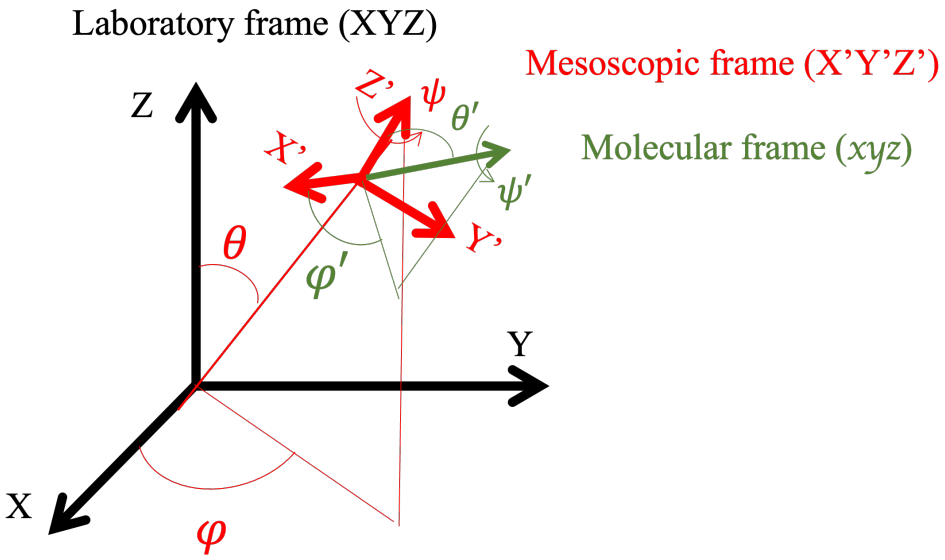


Figure 2.3: Definition of the angle $(\varphi_l, \theta_l, \psi_l)$ and $(\varphi'_l, \theta'_l, \psi'_l)$ which describe the orientation of the scattered objects.

The program reports the same $(a_\Pi^\Theta, b_\Pi^\Theta, c_\Pi^\Theta)$ and $I_2^{\Pi,\Theta}, I_2^{\Pi,\Theta}$ coefficients as in HRS, plus the cross terms b_{V1}, b_{V2}, b_{H1}, b_{H2} that vanish for purely incoherent HRS but may be non-zero for coherent aggregates:

$$I_V(\gamma,\Theta) = a_V^\Theta \cos^4\gamma + b_V^\Theta \cos^2\gamma\sin^2\gamma + c_V^\Theta \sin^4\gamma + b_{1V}^\Theta \cos^3\gamma\sin\gamma + b_{2V}^\Theta \cos\gamma\sin^3\gamma \tag{2.34}$$

$$I_H(\gamma,,\Theta) = a_H^\Theta \cos^4\gamma + b_H^\Theta \cos^2\gamma\sin^2\gamma + c_H^\Theta \sin^4\gamma + b_{1H}^\Theta \cos^3\gamma\sin\gamma + b_{2H}^\Theta \cos\gamma\sin^3\gamma \tag{2.35}$$

with the following values for the different coefficients:

$$a_V^\Theta = \langle \beta_{t,XXX}\beta^*_{t,XXX}\rangle \tag{2.36}$$

$$c_V^\Theta = \langle \beta_{t,XYY}\beta^*_{t,XYY}\rangle \tag{2.37}$$

$$b_V^\Theta = \langle 4\beta_{t,XXY}\beta^*_{t,XXY} + \beta_{t,XXX}\beta^*_{t,XYY} + \beta^*_{t,XXX}\beta_{t,XYY}\rangle \tag{2.38}$$

$$b_{1V}^\Theta = \left\langle 2\beta_{XXX}\beta^*_{XXY} + 2\beta^*_{XXX}\beta_{XXY}\right\rangle \tag{2.39}$$

$$b_{2V}^\Theta = \left\langle 2\beta_{XXY}\beta^*_{XYY} + 2\beta^*_{XXY}\beta_{XYY}\right\rangle \tag{2.40}$$

$$a_H^\Theta = \langle \beta_{t,YXX}\beta^*_{t,YXX}\cos^2\rangle\Theta + \langle \beta_{t,ZXX}\beta^*_{t,ZXX}\rangle \sin^2\Theta - \langle \beta_{t,YXX}\beta^*_{t,ZXX} + \beta_{t,ZXX}\beta^*_{t,YXX}\rangle \cos\Theta\sin\Theta \tag{2.41}$$

$$c_H^\Theta = \langle \beta_{t,YYY}\beta^*_{t,YYY}\cos^2\rangle\Theta + \langle \beta_{t,ZYY}\beta^*_{t,ZYY}\rangle \sin^2\Theta - \langle \beta_{t,YYY}\beta^*_{t,ZYY} + \beta_{t,ZYY}\beta^*_{t,YYY}\rangle \cos\Theta\sin\Theta \tag{2.42}$$

$$\begin{aligned} b_H^\Theta = {} & \left\langle 4\beta_{ZXY}\beta^*_{ZXY} + \beta_{ZXX}\beta^*_{ZYY} + \beta^*_{ZXX}\beta_{ZYY}\right\rangle \sin^2\Theta \\ & + \left\langle 4\beta_{YYX}\beta^*_{YYX} + \beta_{YXX}\beta^*_{YYY} + \beta^*_{YXX}\beta_{YYY}\right\rangle \cos^2\Theta \\ & - \cos\Theta\sin\Theta\big\langle 4\beta_{YYX}\beta^*_{ZXY} + 4\beta_{ZXY}\beta^*_{YYX} + \beta_{ZXX}\beta^*_{YYY} + \beta^*_{ZXX}\beta_{YYY} \\ & \qquad + \beta_{YXX}\beta^*_{ZYY} + \beta_{ZYY}\beta^*_{YXX}\big\rangle \end{aligned} \tag{2.43}$$

$$\begin{aligned} b_{1H}^\Theta = {} & \left\langle 2\beta_{ZXX}\beta^*_{ZXY} + 2\beta^*_{ZXX}\beta_{ZXY}\right\rangle \sin^2\Theta \\ & + \left\langle 2\beta_{YXX}\beta^*_{YYX} + 2\beta^*_{YXX}\beta_{YYX}\right\rangle \cos^2\Theta \\ & - \cos\Theta\sin\Theta\big\langle 2\beta_{ZXX}\beta^*_{ZXY} + 2\beta_{ZXY}\beta^*_{ZXX} + 2\beta_{YXX}\beta^*_{YXY} + 2\beta^*_{YXX}\beta_{YXY}\big\rangle \end{aligned} \tag{2.44}$$

$$\begin{aligned} b_{2H}^\Theta = {} & \left\langle 2\beta_{ZYY}\beta^*_{ZXY} + 2\beta^*_{ZYY}\beta_{ZXY}\right\rangle \sin^2\Theta \\ & + \left\langle 2\beta_{YYY}\beta^*_{YYX} + 2\beta^*_{YYY}\beta_{YYX}\right\rangle \cos^2\Theta \\ & - \cos\Theta\sin\Theta\big\langle 2\beta_{ZYY}\beta^*_{ZXY} + 2\beta_{ZXY}\beta^*_{ZYY} + 2\beta_{YYY}\beta^*_{YXY} + 2\beta^*_{YYY}\beta_{YXY}\big\rangle \end{aligned} \tag{2.45}$$

The polarization-resolved intensities and the ratios $I_2^{\Theta,\Pi}$, $I_4^{\Theta,\Pi}$ follow equations (2.20) to (2.25).

Remark: a coherent calculation with $N = 1$ recovers exactly the HRS numerical result.

2.4 Coherent SHS by a sphere

For colloidal particles the medium is inhomogeneous. Two complex refractive indices are used as defined in the NonLinear Wentzel–Kramers–Brillouin approximation (NLWKB):

- n_p: refractive index of the *particle*,
- n_m: refractive index of the surrounding *medium*.

In the Nonlinear Wentzel–Kramers–Brillouin (NLWKB) approximation the wave-vector magnitudes are

$$k^{\omega} = \frac{2\pi n_p}{\lambda} \qquad \text{(inside the particle)}, \tag{2.46}$$

$$K^{2\omega} = \frac{4\pi n_m}{\lambda} \qquad \text{(in the medium)}. \tag{2.47}$$

(The classical Rayleigh–Gans–Debye limit is recovered by setting $n_p = n_m = n$.)

The intensity is obtained from an effective nonlinear susceptibility tensor $\overset{\leftrightarrow}{\Gamma}{}^{(2)}_{ijk}$ representing the response of the whole particle surface [1]:

$$I_{SHS}(\mathbf{e}^{\omega}_{p-in}, \mathbf{e}^{2\omega}_{p-out}, \mathbf{e}^{2\omega}_{n}) = I_{SHS}(\gamma, \Pi, \Theta) = \mathbf{\Gamma}_{eff}(\gamma, \Pi, \Theta).\mathbf{\Gamma}^{*}_{eff}(\gamma, \Pi, \Theta) \tag{2.48}$$

where $\mathbf{\Gamma}_{eff}$ is defined as:

$$\mathbf{\Gamma}_{eff} = \mathbf{e}^{2\omega}_{n} \times (\overset{\leftrightarrow}{\mathbf{\Gamma}}{}^{\mathbf{(2)}} : \mathbf{e}^{\omega}_{p-in}\mathbf{e}^{\omega}_{p-in}) \times \mathbf{e}^{2\omega}_{n} \tag{2.49}$$

here

$$\overset{\leftrightarrow}{\mathbf{\Gamma}}{}^{\mathbf{(2)}} = \iint_{surface} \overset{\leftrightarrow}{\chi}{}^{(2)} e^{i\Delta\mathbf{k}.\mathbf{r}} dS \tag{2.50}$$

where $\overset{\leftrightarrow}{\chi}{}^{(2)}$ is the surface second order nonlinear susceptibility tensor expressed in the particle frame (x',y',z').

For a sphere of radius R,

$$\Gamma^{(2)}_{i'j'k'}(\Theta) = \int_0^{2\pi} \mathrm{d}\varphi_c \int_0^{\pi} \mathrm{d}\theta_c \, R^2 \sin\theta_c \, \chi^{(2)}_{i'j'k'}(\varphi_c, \theta_c) \, e^{i\,\Phi(\varphi_c,\theta_c,\Theta)}, \tag{2.51}$$

where

$$\varphi(\phi_c, \theta_c, \Theta) = (-2k^{\omega} + K^{2\omega}\cos\Theta)\, R\cos\theta_c + K^{2\omega}\sin\Theta\, R\sin\theta_c\sin\phi_c \tag{2.52}$$

accounts for optical path differences.

Here $\chi^{(2)}_{i'j'k'}$ is the surface susceptibility expressed in the particle frame which can be deduced from the surface susceptibility expressed in the surface frame by:

$$\chi^{(2)}_{i'j'k'}(\varphi_c, \theta_c) = \sum_{i=1}^{3}\sum_{j=1}^{3}\sum_{k=1}^{3} T_{i'i}(\varphi_c, \theta_c, 0) T_{j'j}(\varphi_c, \theta_c, 0) T_{k'k}(\varphi_c, \theta_c, 0)\, \chi^{(2)}_{ijk}, \tag{2.53}$$

Because a sphere is isotropic, no further orientational average is required; that is why in equation (2.64) no $< . >$ is needed. In other term, one can considered that $\Gamma^{(2)}_{i'j'k'}$ is equal to $\Gamma^{(2)}_{IJK}$. The $(a_V, b_V, c_V, a_H, b_H, c_H)$ coefficients and I_2, I_4 ratios as determined by:

$$a^{\Theta}_{V} = \Gamma_{XXX}\Gamma^{*}_{XXX} \tag{2.54}$$

$$c^{\Theta}_{V} = \Gamma_{XYY}\Gamma^{*}_{XYY} \tag{2.55}$$

$$b^{\Theta}_{V} = 4\Gamma_{XXY}\Gamma^{*}_{XXY} + \Gamma_{XXX}\Gamma^{*}_{XYY} + \Gamma^{*}_{XXX}\Gamma_{XYY} \tag{2.56}$$

$$a_H^\Theta = \Gamma_{YXX}\Gamma^*_{YXX}\cos^2\Theta + \Gamma_{ZXX}\Gamma^*_{ZXX}\sin^2\Theta - (\Gamma_{YXX}\Gamma^*_{ZXX} + \Gamma_{ZXX}\Gamma^*_{YXX})\cos\Theta\sin\Theta \quad (2.57)$$

$$c_H^\Theta = \Gamma_{YYY}\Gamma^*_{YYY}\cos^2\Theta + \Gamma_{ZYY}\Gamma^*_{ZYY}\sin^2\Theta - (\Gamma_{YYY}\Gamma^*_{ZYY} + \Gamma_{ZYY}\Gamma^*_{YYY})\cos\Theta\sin\Theta \quad (2.58)$$

$$\begin{aligned} b_H^\Theta = {} & (4\Gamma_{ZXY}\Gamma^*_{ZXY} + \Gamma_{ZXX}\Gamma^*_{ZYY} + \Gamma^*_{ZXX}\Gamma_{ZYY})\sin^2\Theta \\ & + (4\Gamma_{YYX}\Gamma^*_{YYX} + \Gamma_{YXX}\Gamma^*_{YYY} + \Gamma^*_{YXX}\Gamma_{YYY})\cos^2\Theta \\ & - \cos\Theta\sin\Theta(4\Gamma_{YYX}\Gamma^*_{ZXY} + 4\Gamma_{ZXY}\Gamma^*_{YYX} + \Gamma_{ZXX}\Gamma^*_{YYY} + \Gamma^*_{ZXX}\Gamma_{YYY} \\ & \quad + \Gamma_{YXX}\Gamma^*_{ZYY} + \Gamma_{ZYY}\Gamma^*_{YXX}) \end{aligned} \quad (2.59)$$

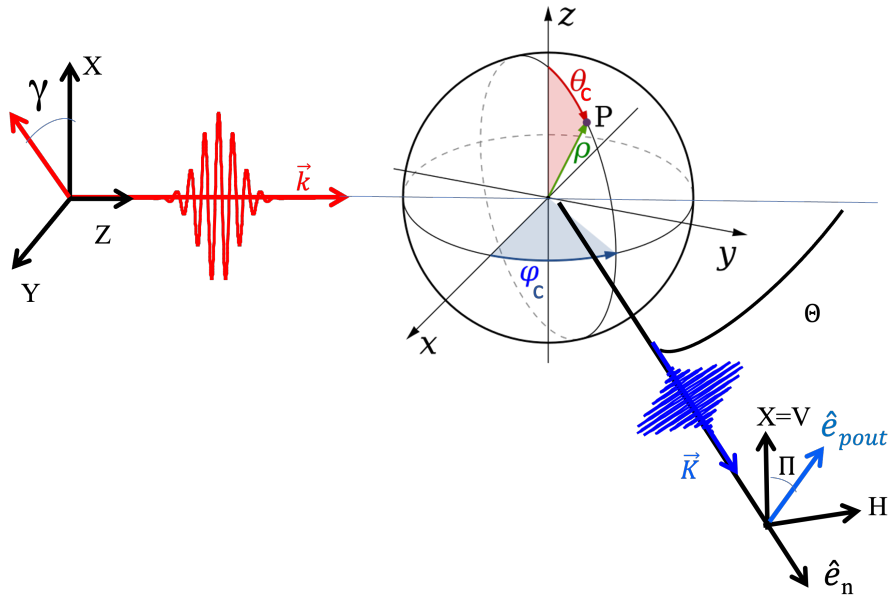


Figure 2.4: Definition of the angle (φ_c, θ_c) which describe the position on the surface sphere.

2.5 Coherent SHS by a cylinder

2.5.1 Physical model

As was the case for the sphere module, for the cylinder, the surrounding medium and the particle interior are optically distinct. Two complex refractive indices enter the calculation:

- n_p: refractive index of the *particle* (cylinder material),
- n_m: refractive index of the surrounding *medium*.

The wave-vector magnitudes are

$$k_\omega = \frac{2\pi n_p}{\lambda} \qquad \text{(fundamental, inside the particle)}, \quad (2.60)$$

$$k_{2\omega} = \frac{4\pi n_m}{\lambda} \qquad \text{(harmonic, in the medium)}. \quad (2.61)$$

2.5.2 Geometry and local surface frame

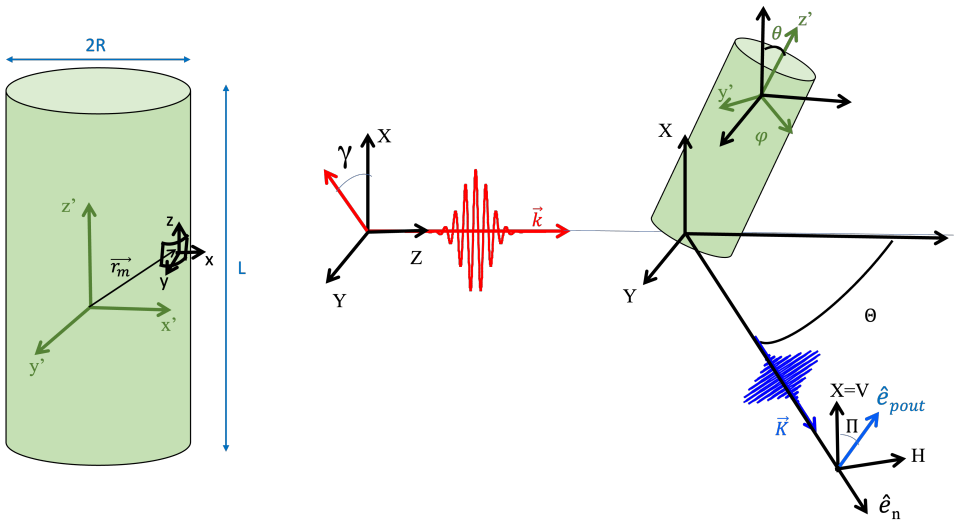


Figure 2.5: Cylinder of radius R and length L. A surface element is labeled by the azimuthal angle $\theta_c \in [0, 2\pi]$ and the axial coordinate $z_c \in [-L/2, L/2]$. The local surface frame is labeled by (x,y,z) , the cylinder frame by (x',y',z') and it is oriented in the laboratory frame (X,Y,Z) by Euler angles (ϕ, θ, ψ).

A point on the lateral surface has mesoscopic Cartesian coordinates

$$x' = R\cos\theta_c, \qquad y' = R\sin\theta_c, \qquad z' = z_c. \tag{2.62}$$

Its laboratory coordinates are obtained with the ZYZ rotation matrix $\mathbf{T}(\varphi, \theta, \psi)$:

$$r_i^{\text{lab}}(\varphi,\theta,\psi) = T_{i0}(\varphi,\theta,\psi)\,x' + T_{i1}(\varphi,\theta,\psi)\,y' + T_{i2}(\varphi,\theta,\psi)\,z', \qquad i \in \{x,y,z\}. \tag{2.63}$$

2.5.3 Effective nonlinear susceptibility

The scattered intensity is computed from an effective susceptibility that integrates the local response over the whole lateral surface [1]:

$$I_{SHS}(\mathbf{e}^{\omega}_{p-in}, \mathbf{e}^{2\omega}_{p-out}, \mathbf{e}^{2\omega}_{n}) = I_{SHS}(\gamma, \Pi, \Theta) =< \mathbf{\Gamma}_{eff}(\gamma,\Pi,\Theta).\mathbf{\Gamma}^{*}_{eff}(\gamma,\Pi,\Theta) > \tag{2.64}$$

Note that contrary to the sphere, the average over all cylinder orientation $< \ . \ >$ is here necessary as the cylinder is not isotropic. $\mathbf{\Gamma}_{eff}$ is defined by:

$$\mathbf{\Gamma}_{eff} = \mathbf{e}^{2\omega}_{n} \times (\overset{\leftrightarrow}{\mathbf{\Gamma}^{(\mathbf{2})}} : \mathbf{e}^{\omega}_{p-in}\mathbf{e}^{\omega}_{p-in}) \times \mathbf{e}^{2\omega}_{n} \tag{2.65}$$

and

$$\overset{\leftrightarrow}{\mathbf{\Gamma}^{(\mathbf{2})}} = \iint_{surface} \overset{\leftrightarrow}{\chi^{(2)}} e^{i\Delta\mathbf{k}.\mathbf{r}} dS \tag{2.66}$$

Each Cartesian component of $\overset{\leftrightarrow}{\mathbf{\Gamma}^{(\mathbf{2})}}$ is

$$\overset{\leftrightarrow(2)}{\Gamma}_{IJK}(\varphi,\theta,\psi,\Theta) = \int_0^{2\pi} \mathrm{d}\theta_c \int_{-L/2}^{L/2} \mathrm{d}z_c \; R \overset{\leftrightarrow(2)}{\chi}_{IJK}(\theta_c, z_c, \varphi, \theta, \psi)\, e^{\,i\,\Phi(\theta_c, z_c, \Theta, k^{\omega}, K^{2\omega})}, \tag{2.67}$$

where the NLWKB phase collected along the optical path is

$$\Phi(\theta_c, z_c, \Theta, k^{\omega}, K^{2\omega}) = \left(-2k^{\omega} + K^{2\omega}\cos\Theta\right) r_z^{\text{lab}} + K^{2\omega}\sin\Theta\, r_y^{\text{lab}}. \tag{2.68}$$

The laboratory components r_y^{lab} and r_z^{lab} are those of (2.63).

The $\overset{\leftrightarrow}{\chi}^{(2)}_{IJK}$ tensor elements expressed in the laboratory frame are deduced according to:

$$\overset{\leftrightarrow}{\chi}^{(2)}_{IJK}(\theta_c, z_c, \varphi, \theta, \psi) = \sum_{i=1'}^{3}\sum_{j=1'}^{3}\sum_{k'=1}^{3} T_{Ii'}(\varphi,\theta,\psi)T_{Jj'}(\varphi,\theta,\psi)T_{Kk'}(\varphi,\theta,\psi)\,\overset{\leftrightarrow}{\chi}^{(2)}_{i'j'k'}(\theta_c, z_c) \tag{2.69}$$

The i second-order susceptibility tensor $\chi^{(2)}_{i'j'k'}$ expressed in the cylinder frame is determined by:

$$\overset{\leftrightarrow}{\chi}^{(2)}_{i'j'k'}(\theta_c) = \sum_{i=1}^{3}\sum_{j=1}^{3}\sum_{k=1}^{3} T_{i'i}(\theta_c, \frac{\pi}{2}, 0)T_{j'j}(\theta_c, \frac{\pi}{2}, 0)T_{k'k}(\theta_c, \frac{\pi}{2}, 0)\,\overset{\leftrightarrow}{\chi}^{(2)}_{ijk} \tag{2.70}$$

where $\overset{\leftrightarrow}{\chi}^{(2)}_{ijk}$ is the second order susceptibility tensor expressed in the local surface frame.

2.5.4 Numerical quadrature and orientational average

The double integral (2.67) is evaluated by a mid-point quadrature on a uniform grid. For an isotropic suspension of cylinders an additional average over the orientation of the cylinder axis (ϕ, θ) is required:

$$\langle \Gamma^{(2)}_{IJK}(\Theta)\rangle = \frac{1}{4\pi}\int_0^{2\pi} \mathrm{d}\varphi \int_0^{\pi} \mathrm{d}\theta\, \sin\theta\, \Gamma^{(2)}_{IJK}(\varphi, \theta, \psi = 0, \Theta). \tag{2.71}$$

2.5.5 Polarization-resolved coefficients

From the averaged laboratory components the six coefficients can be deduced :

$$a_V^{\Theta} =< \Gamma_{XXX}\Gamma^*_{XXX} > \tag{2.72}$$

$$c_V^{\Theta} =< \Gamma_{XYY}\Gamma^*_{XYY} > \tag{2.73}$$

$$b_V^{\Theta} =< 4\Gamma_{XXY}\Gamma^*_{XXY} + \Gamma_{XXX}\Gamma^*_{XYY} + \Gamma^*_{XXX}\Gamma_{XYY} > \tag{2.74}$$

$$a_H^{\Theta} =< \Gamma_{YXX}\Gamma^*_{YXX} > \cos^2\Theta + < \Gamma_{ZXX}\Gamma^*_{ZXX} > \sin^2\Theta - < \Gamma_{YXX}\Gamma^*_{ZXX} + \Gamma_{ZXX}\Gamma^*_{YXX} > \cos\Theta\sin\Theta \tag{2.75}$$

$$c_H^{\Theta} =< \Gamma_{YYY}\Gamma^*_{YYY} > \cos^2\Theta + < \Gamma_{ZYY}\Gamma^*_{ZYY} > \sin^2\Theta - < \Gamma_{YYY}\Gamma^*_{ZYY} + \Gamma_{ZYY}\Gamma^*_{YYY} > \cos\Theta\sin\Theta \tag{2.76}$$

$$\begin{aligned} b_H^{\Theta} =& < 4\Gamma_{ZXY}\Gamma^*_{ZXY} + \Gamma_{ZXX}\Gamma^*_{ZYY} + \Gamma^*_{ZXX}\Gamma_{ZYY} > \sin^2\Theta \\ &+ < 4\Gamma_{YYX}\Gamma^*_{YYX} + \Gamma_{YXX}\Gamma^*_{YYY} + \Gamma^*_{YXX}\Gamma_{YYY} > \cos^2\Theta \\ &- \cos\Theta\sin\Theta < 4\Gamma_{YYX}\Gamma^*_{ZXY} + 4\Gamma_{ZXY}\Gamma^*_{YYX} + \Gamma_{ZXX}\Gamma^*_{YYY} + \Gamma^*_{ZXX}\Gamma_{YYY} \\ &\quad + \Gamma_{YXX}\Gamma^*_{ZYY} + \Gamma_{ZYY}\Gamma^*_{YXX} > \end{aligned} \tag{2.77}$$

The polarization-resolved intensities and the ratios $I_2^{\Theta,\Pi}$, $I_4^{\Theta,\Pi}$ also follow equations (2.20) to (2.25).

Chapter 3

HRS program

3.1 Inputs and outputs

The only molecular input is the hyperpolarizability tensor β_{ijk} in atomic units, provided in an `input_beta` file:

```
# comment line
xxx  0.0
xxy  0.0
xxz  0.0
...
zzz  value
```

(27 components, one per line after the header, label then value).

Outputs: coefficients $a_V, b_V, c_V, a_H, b_H, c_H$, ratios $I_{2V}, I_{4V}, I_{2H}, I_{4H}$, and the file `out_plot` for polarization or angular curves.

3.2 Command line

```
./bin/HRS <keyword> <input_beta> <output_file>
```

Plotting:

```
python3 Src/graph_polar.py Work/out_plot
python3 Src/graph_angle.py Work/out_plot
```

Chapter 4

SHS program

4.1 Inputs

1. Hyperpolarizability tensor: same `input_beta` format as HRS.
2. Position/orientation file (mesoscopic frame): six columns per line $(\phi, \theta, \psi, x, y, z)$ for each dipole. **Angles in radians.** The number of lines is the number of molecules N.
3. Interactive parameters: N, wavelength λ (nm), real and imaginary parts of n, scattering angles / number of points.

Generation scripts in `Src/` can build orientation files for spheres, spheroids, cylinders, etc. (see Chapter 8).

4.2 Command line

```
./bin/SHS <keyword> <input_beta> <input_orientation> <output_file>
```

Plotting:

```
python3 Src/graph_polar.py    Work/out_plot
python3 Src/graph_angle.py    Work/out_plot
python3 Src/graph_assembly.py Work/orientation_file
```

Chapter 5

Sphere Object (`sphere_SHS`)

5.1 Inputs

- Surface susceptibility / hyperpolarizability tensor (`input_beta`, same format).
- wavelength λ (nm),
- n_p real and imaginary parts (particle),
- n_m real and imaginary parts (medium),
- sphere radius R (nm),
- scattering angle(s) or number of angular points.

5.2 Command line

```
./bin/sphere_SHS <keyword> <input_beta> <output_file>
```

Chapter 6

Cylinder Object (`cylinder_SHS`)

6.1 Inputs

- Surface susceptibility tensor (`input_beta`).
- wavelength λ (nm),
- n_p (real, imag), n_m (real, imag),
- cylinder radius R (nm) and length L (nm),
- scattering angle(s) or number of points.

6.2 Command line

```
./bin/cylinder_SHS <keyword> <input_beta> <output_file>
```

Chapter 7

Graphical User Interface

7.1 Launch

```
python3 run_gui.py
```

On first start the GUI compiles (or updates) the four C modules into `bin/`.

7.2 Common layout

Four tabs: **HRS**, **SHS**, **Sphere Object**, **Cylinder Object**.
Each tab provides:

- calculation type (keyword),
- numerical parameters,
- tensor (and orientation) editors with Load / Load example / Load Gaussian (HRS),
- **Run** button (C sources are compiled automatically at startup),
- live log (stdout/stderr of the C program),
- polarization or angular plots, with export buttons.

7.3 HRS tab

- Keyword: `polarplot_single` / `polarplot_integrate` / `angle_scattering`.
- Scattering angle(s) or number of points.
- Hyperpolarizability editor.
- **Load Gaussian file...**: runs `script_gauss2PySHS.py` on a Gaussian `.log` and fills the tensor.

7.4 SHS tab

- Keyword and angle parameters as above.
- Number of dipoles N, wavelength, n (real, imag).
- Hyperpolarizability and position/orientation editors.

- Dropdown to choose a generation script (sphere, sphere_random, spheroid_random, cylinder_random, or custom script) and **Generate positions/orientations** button.
- After a run: intensity plot and dipole-assembly plot; export buttons for both.

Note: N is synchronized with the number of lines of the orientation file when possible, to avoid scaling inconsistencies.

7.5 Sphere Object and Cylinder Object tabs

- Keyword and angle parameters.
- Wavelength.
- n_p **(particle)** real and imaginary parts.
- n_m **(medium)** real and imaginary parts.
- Radius (and length for the cylinder).
- Surface $\chi^{(2)}$ / hyperpolarizability editor.

NLWKB is used by default (two refractive indices). Setting $n_p = n_m$ recovers the classical single-index limit.

7.6 Results and export

- Live progress messages appear in the Results pane.
- Polar plots: $I_V(\gamma)$, $I_H(\gamma)$.
- Angular plots: IH(0°), IH(90°), IV(0°), IV(90°) vs Θ.
- SHS: additional 3-D assembly view of dipoles.
- Export buttons save figures (PNG/PDF) on demand.

Chapter 8

Additional scripts

8.1 Orientation / position generators (SHS)

All scripts write a six-column file (angles in radians, positions in the chosen unit):

```
python3 Src/script_sphere.py          Work/out_orient
python3 Src/script_sphere_random.py   Work/out_orient
python3 Src/script_spheroid_random.py Work/out_orient
python3 Src/script_cylinder_random.py Work/out_orient
```

Typical prompts: radius (and axes for spheroid / length for cylinder), number of dipoles, optional Gaussian angular disorder σ.

8.2 Gaussian → PySHS tensor

```
python3 Src/script_gauss2PySHS.py  path/to/job.log  Work/beta_from_gauss
```

Extracts the static or dynamic hyperpolarizability from a Gaussian output and writes a PySHS `input_beta` file. Accessible from the HRS tab via **Load Gaussian file...**.

8.3 Plotting

```
python3 Src/graph_polar.py    Work/out_plot
python3 Src/graph_angle.py    Work/out_plot
python3 Src/graph_assembly.py Work/orientation_file
```

Bibliography

[1] S. Roke, M. Bonn and A. V. Petukhov, "Nonlinear optical scattering: The concept of effective susceptibility," *Phys. Rev. B* **70**, 115106 (2004).

Acknowledgements. Support was provided by the CAMOMILS project, ANR-15-CE21-0002. Contributors: Dr Pierre-Marie Gassin, Dr Lotfi Boudjema, Dr Gaelle Gassin.